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The role of ore-forming conditions in high germanium enrichment of colloform sphalerite: A case study of the Banbianjie deposit, SW China

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ABSTRACT

Colloform sulfides host significant germanium (Ge) resources in South China. The Banbianjie deposit, situated in southwest China, is a carbonate-hosted Zn-Ge deposit (0.8 Mt. @ 1.78 to 9.50 % Zn and > 800 t @ 100–110 ppm Ge) and is rich in colloform sphalerite with widespread occurrence of Ge-rich nanoparticles. Previous studies documented that Ge enrichment in sphalerite may be correlated to internal mechanisms, however, the impact of external factors, such as source and fluid mixing, on Ge enrichment is poorly understood. In this study, we applied in situ trace elements and S-Pb isotopic compositions to interpret the external controls influencing Ge distribution. New laser-ablation inductively coupled mass spectroscopy (LA-ICP-MS) analyses reveal that sphalerite contains very heterogeneous Ge contents from 221 to 1916 ppm (mean = 843 ppm Ge), positively correlating with Fe, Mn, and Pb and negatively with Cd concentrations. Sulfur isotopic compositions infer a predominant sulfur source (~ − 10 ‰) and a subordinate sulfur source (~ − 1 ‰). The Pb isotopic ratios of sphalerite plot on the upper crust curve, indicating a crustal source for the metals. The sulfur isotope geothermometer estimates the formation temperatures of sphalerite mainly below 250 °C and pH changes may cause the transition of sulfide phases. Trace elements and $\delta^{34}S$ values indicate a predominant Ge-rich fluid and a subordinate Cd-rich fluid. The influxes of Cd-rich fluid are correlated with local Ge-poor zonings in sphalerite. We propose that fluid mixing can locally influence the Ge contents in sphalerite.

1. Introduction

Germanium (Ge) is considered a critical metal due to its key applications and high supply risk (Gulley et al., 2018; Tao et al., 2019; Wang, 2019; Wei et al., 2019; Zhai et al., 2019; Cugerone et al., 2020; Zhou et al., 2021a; Kelley et al., 2021). Ge is used in infrared goggles (Gulley et al., 2018), high-performance fibre-optic cables, and semiconductor materials (Frenzel et al., 2014). Ge is mainly extracted as a by-product of coal (Zhuang et al., 2006; Du et al., 2009; Hu et al., 2009; Qi et al., 2011) or base metal deposits (e.g., carbonate-hosted, sedimentary-exhalative, and volcanogenic massive sulfides; Frenzel et al., 2014, 2017; Paradis, 2015; Lawley et al., 2022; Luo et al., 2022; Liu et al., 2023; Torró et al., 2023).

In the base metal deposits, Ge is dominantly hosted in the sphalerite, especially the low-temperature sphalerite (like colloform sphalerite, Höll et al., 2007; Barrie et al., 2009; Cook et al., 2009; Ye et al., 2011; Torró et al., 2023). The colloform sphalerite represents the sphalerite with colloform textures that involve the aggregation of microcrystals and show rounded or layered structures (Barrie et al., 2009; Pfaff et al., 2011). Benites et al. (2021) reported that colloform sphalerite shows relatively higher Ge contents than other crystalline sphalerites in the Morococha district, Peru. Luo et al. (2022) documented the high Ge contents (up to 1553 ppm) in colloform sphalerite in SW China. Although considerable research has been studying the formation of colloform sphalerite (Barrie et al., 2009; Pfaff et al., 2011), there is still confusion about the enrichment of Ge in it.

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The Banbianjie carbonate-hosted deposit contains resources of 0.8 Mt. @ 1.78 to 9.5 % Zn and > 800 t @ 100–110 ppm Ge (An et al., 2022; Sun et al., 2024). The distinctive characteristic of the Banbianjie deposit lies in the occurrence of Ge as nanoscale particles within sphalerite (Sun et al., 2024). Sun et al. (2024) discussed the internal controls on the Ge enrichment in sphalerite and concluded on the presence of self-organization processes. However, the external ore-forming environment (sources, ore temperatures, pH, redox state, or fluid compositions) has not been reported.

In this study, we describe the deposit geology and mineralogy of the Banbianjie deposit, reveal the distribution of Ge in sphalerite with its relationship with S-Pb isotopic compositions, determine the sources of ore-forming materials, and discuss the influence of external mechanisms (sources of ore-forming materials, temperatures, pH, redox state, and fluid compositions). Finally, we propose that fluid mixing may cause the local Ge-poor oscillatory zonings, whereas the effect of temperatures and redox state may be limited.

2. Regional geology

The Banbianjie deposit is located in the southeast margin of the Yangtze block (Fig. 1a), which is composed of Late-Proterozoic metamorphic and metasedimentary rocks, and Cambrian to Permian-Triassic sedimentary rocks (Fig. 1b). The Proterozoic metamorphic rocks, mainly in the eastern part, include greywacke, slate, and schist, experiencing low-grade greenschist-facies metamorphism (Wang et al., 2010). The Cambrian to Ordovician succession consists of deep-sea chert and black shale in the lower part and transitions to dolomite, muddy limestone, wackestone, and arenaceous shale in the upper part. The Silurian comprises arenaceous shale and sandstone with minor limestone (Yao and Li, 2016). Middle-Devonian sandstone lies unconformably on Silurian arenaceous shales (Yao and Li, 2016). The upper Devonian rocks are constituted of argillaceous dolomite, dolomite with vug, dolomite, and limestone. The Carboniferous rocks are composed of limestone, wackestone, with some carbonaceous shale, coal, and sandstone. The lower Permian consists of quartz sandstone along with coal and claystone, whereas the upper Permian is characterized by chert limestone, dolomitic limestone, and dolomite. The Triassic comprises gray wackestone, muddy limestone, and calcareous shale in the lower part and the rhythmical interlayers of sandstone and siltstone in the upper part (Yao and Li, 2016).

The regional tectonic framework is dominated by N-S, E-W, and NE-NW-trending structures (Fig. 1b). The N-S striking structures are mainly composed of box anticlines (e.g., Huangsi and Wangsi anticlines) and

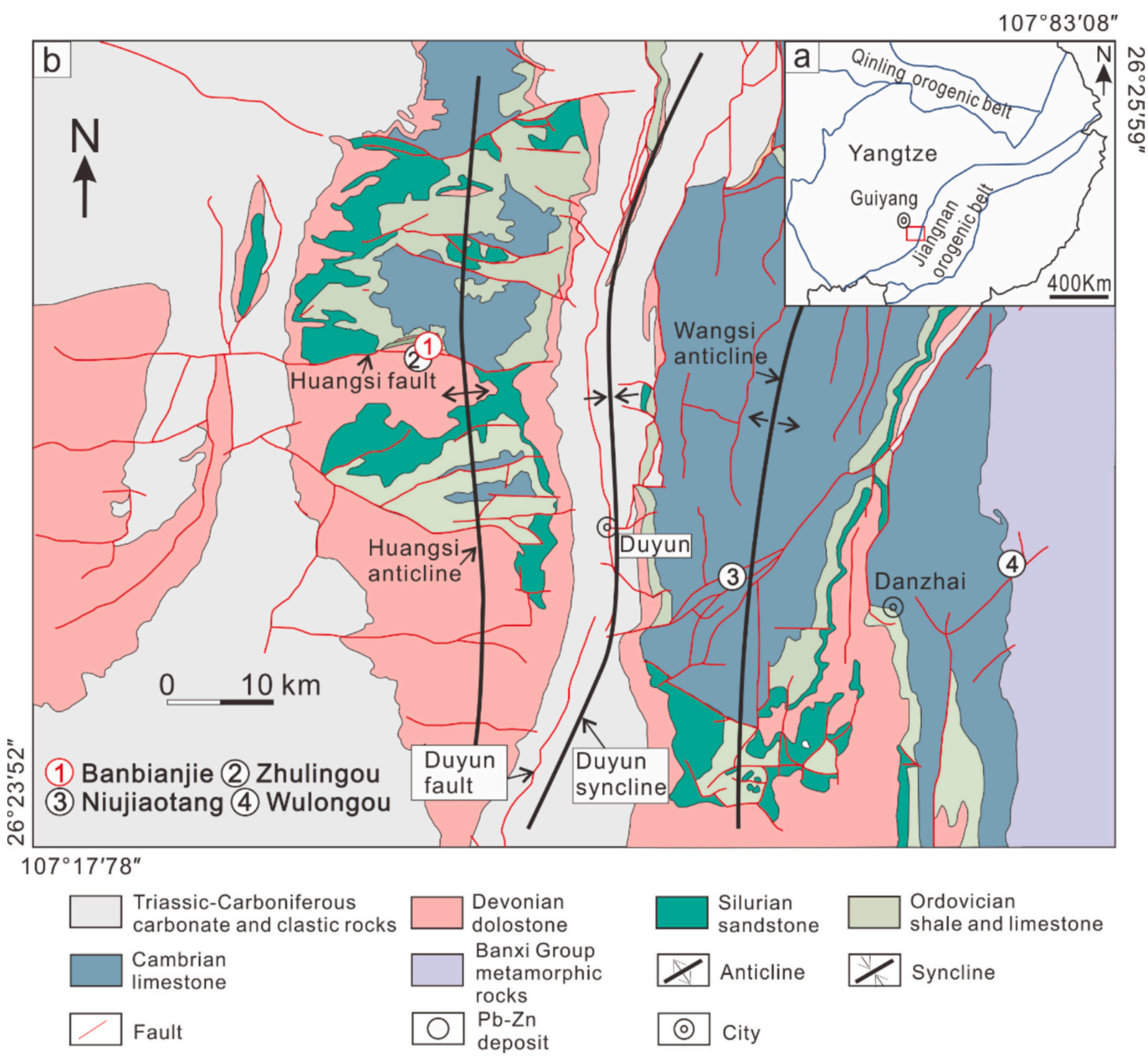

Fig. 1. Geological map of the southeastern margin of the Yangtze Block, South China (modified from Sun et al., 2024). (a) Location of the southeastern margin of the Yangtze Block. (b) Geological map of SE Yangtze Block.

chevron synclines (e.g., Duyun syncline; Fig. 1b). These structures are frequently cut by N-S striking thrust faults (Fig. 1b). The Wangsi anticline is cut by N-S and NE-SW striking normal faults, whereas the Huangsi anticline is cut by E-W striking normal faults (e.g., Huangsi fault) (Fig. 1b).

The N-S Duyun thrust fault and the E-W Huangsi normal fault are considered to have initially formed during the Devonian orogeny (Chen et al., 2006). However, the Mesozoic orogeny resulted in the formation of similar N-S striking folds and reactivation of preexisting faults, and possibly the NE-SW and some of the E-W striking faults (GBGMR (Guizhou bureau of geology and mineral resources), 1987).

3. The Banbianjie deposit

3.1. Deposit geology

The strata consist of Silurian to Middle-Devonian clastic rocks, upper Devonian dolostone, Carboniferous carbonate rocks, and Permian clastic rocks (Chen et al., 2006). The ore bodies are hosted in the Devonian Gaopochang Formation (Fig. 2a), which shows a variable thickness between 380 and 1074 m. The lower part of the Gaopochang Formation consists of dark-gray, thick, fine-grained dolostone and argillaceous dolostone (Fig. 2b-c). The middle part is characterized by dark-gray, thick bioclastic dolostone interbedded with some black, grayish-yellow, or grayish-green mudstone (Fig. 2b-c). A characteristic feature of the dolostone in this part is an abundance of dolomite geodes (Fig. 3a-b), with a 3–20 cm size. The upper part of this formation is dominantly composed of thick light-gray or gray fine-grained dolostone, with a small portion of argillaceous dolostone. The Zn-Ge mineralization mainly occurs in the middle part of the Gaopochang Formation (Fig. 2b-c).

The major structure in the deposit is the *E*-W striking Banbianjie anticline and the E-W striking Huangsi fault zone, located on the south limb of this anticline (Chen et al., 2006, Fig. 2a). The Huangsi fault zone comprises a set of E-W striking faults with a general dip at 40–70°S. Chen et al. (2006) suggested that the fault zone is *syn*-sedimentary, controlling the lithofacies of the Gaopochang Formation, but the Carboniferous-Triassic rocks are crosscut by the Huangsi fault, which indicates that the Huangsi fault was, at least, reactivated after Carboniferous ages.

3.2. Ore bodies

The Zn-Ge mineralization, hosted in the middle part of the Gaopochang dolostone, is essentially composed of six stratabound and lenticular ore bodies. Mineralization is hosted in bedded dark-gray dolostone with a fine-crystalline texture (Fig. 3a-b). The No. 2 ore body, the largest one in Banbianjie, has a thickness ranging from 1.13 to 11.8 m and contains ores with Zn grades between 1.78 and 9.5 % and Ge grades of ~ 100 g/t. The ore bodies are composed of multiple Zn-rich veins that are parallel to or crosscut the bedding of the dark-gray dolostone (Fig. 3a-b). These veins vary in thickness (1–50 cm) and locally host dissolution breccias (Fig. 3a-c). Sulfides in the veins show macroscopic oscillatory zoning patterns (Fig. 3d) and colloform texture (Fig. 3d-f).

4. Sampling and analysis methods

Samples from the Banbianjie deposit were collected from underground galleries and drill cores. Forty-five thin sections were studied with optical microscopy and scanning electron microscopy (SEM). Among them, sphalerite and pyrite in samples BBJ-1, BBJ-5, and ZK1008 were measured with laser-inductively coupled plasma-mass spectrometry (LA-ICP-MS) and LA-multiple collector (MC)-ICP-MS spot analyses for trace element (Wilson et al., 2002) and S-Pb isotopic compositions, respectively. Trace element analysis of sphalerite was performed by LA-ICP-MS at Guangzhou Tuoyan Analytical Technology Co., Ltd., located

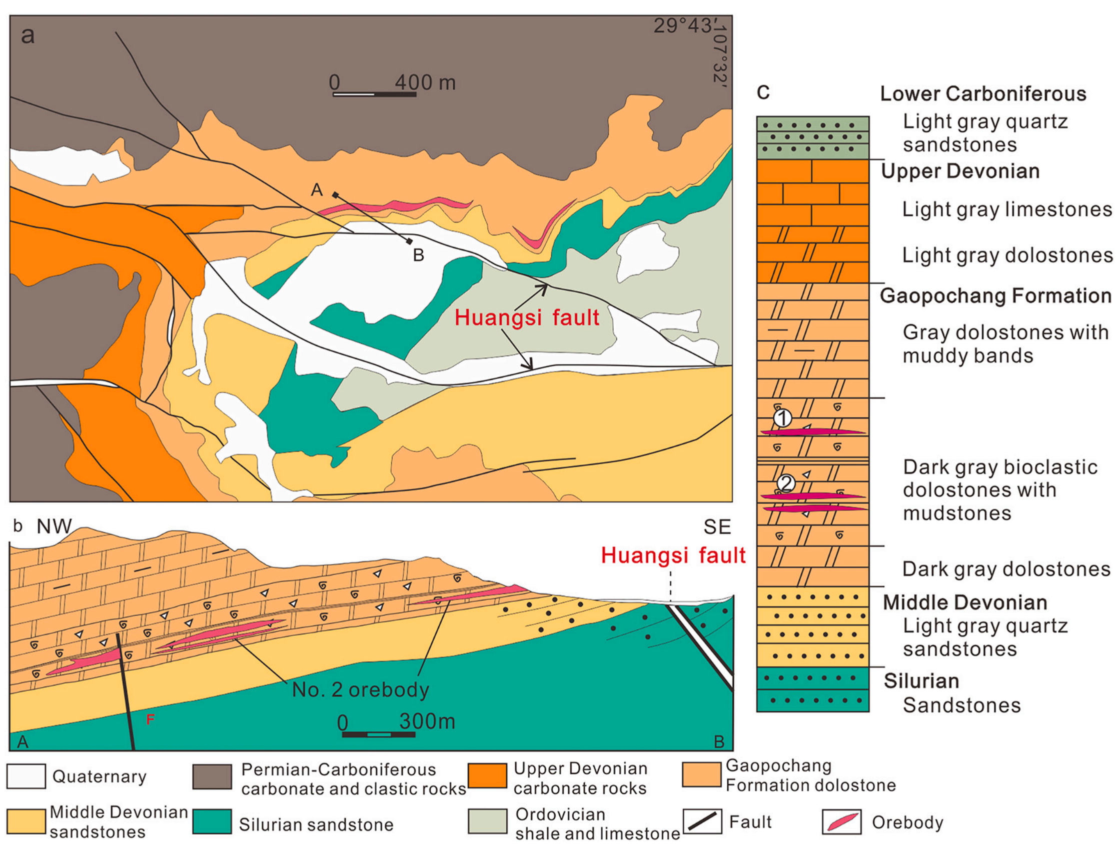

Fig. 2. Geological map of the Banbianjie deposit (modified from Sun et al., 2024). (a) Geological map. (b) Cross-section map. (c) Stratigraphic log.

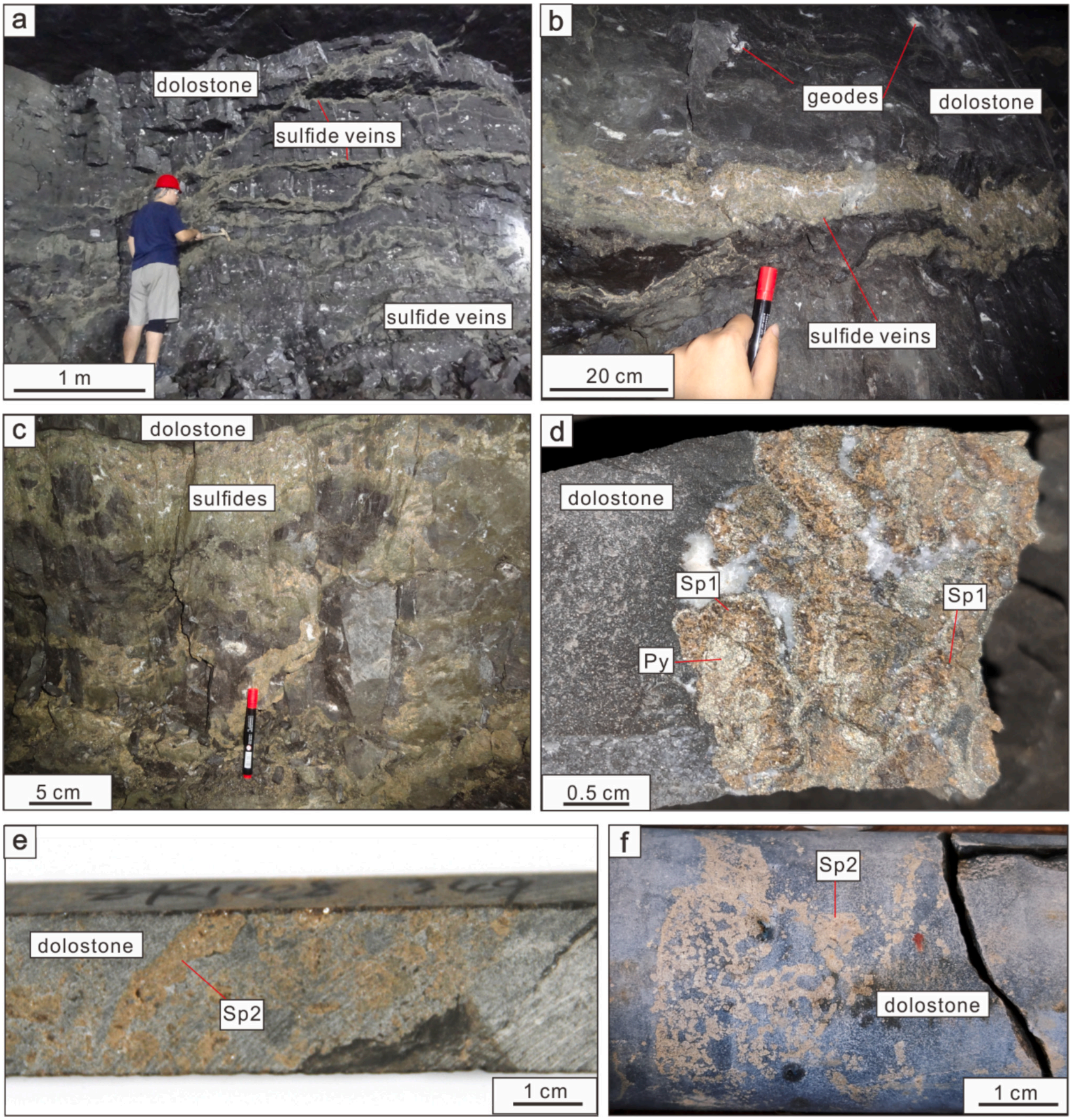

Fig. 3. Field photographs of the Zn-Ge mineralization in the Banbianjie deposit. (a) Stratabound sulfide veins along beddings (white lines) and crosscutting veins (red color). (b) Stratabound sulfide veins in dolostone with geodes. (c) Stratabound sulfide veins with locally crosscutting veins (red color). (d) Sulfide veins consisting of sphalerite, pyrite, and dolomite. (e) Sp2 veins and disseminated Sp2 grains in dolostone. (f) Lumpy aggregates of Sp2 in dolostone. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

in Guangzhou, China. Sulfur isotopic ratio analysis of sulfides was conducted at Nanjing FocuMS Technology Co. Ltd., situated in Nanjing, China. Lead isotope analysis of sphalerite was carried out at Wuhan Sample Solution Analytical Technology Co., Ltd., located in Wuhan, China. Detailed descriptions are shown in the ESM 1.

5. Results

5.1. Mineralogy

The Banbianjie deposit exhibits two distinct types of ores. Type 1 ore predominantly comprises red- to brown sphalerite (Sp1) (Fig. 3d), pyrite (Py1), and acicular marcasite with minor calcite. The Sp1a and Py1a texture is characterized by disseminated and subhedral grains within the dolostone host (Fig. 4b). Sp1b is composed of colloform sphalerite, and associated Py1b crystals form layers, sectors, or globular textures within the vein (Fig. 4a, c). The thicknesses of both the individual sphalerite and pyrite layers vary from 1 to 5 mm (Fig. 4a). Under transmitted light, colloform Sp1b exhibits oscillatory color zonings, with brown to black zonings composed of acicular Sp1b crystals oriented perpendicular to the color zonings (Fig. 4c, d). However, in some cases, light-yellow color zonings may appear, mainly consisting of euhedral sphalerite crystals

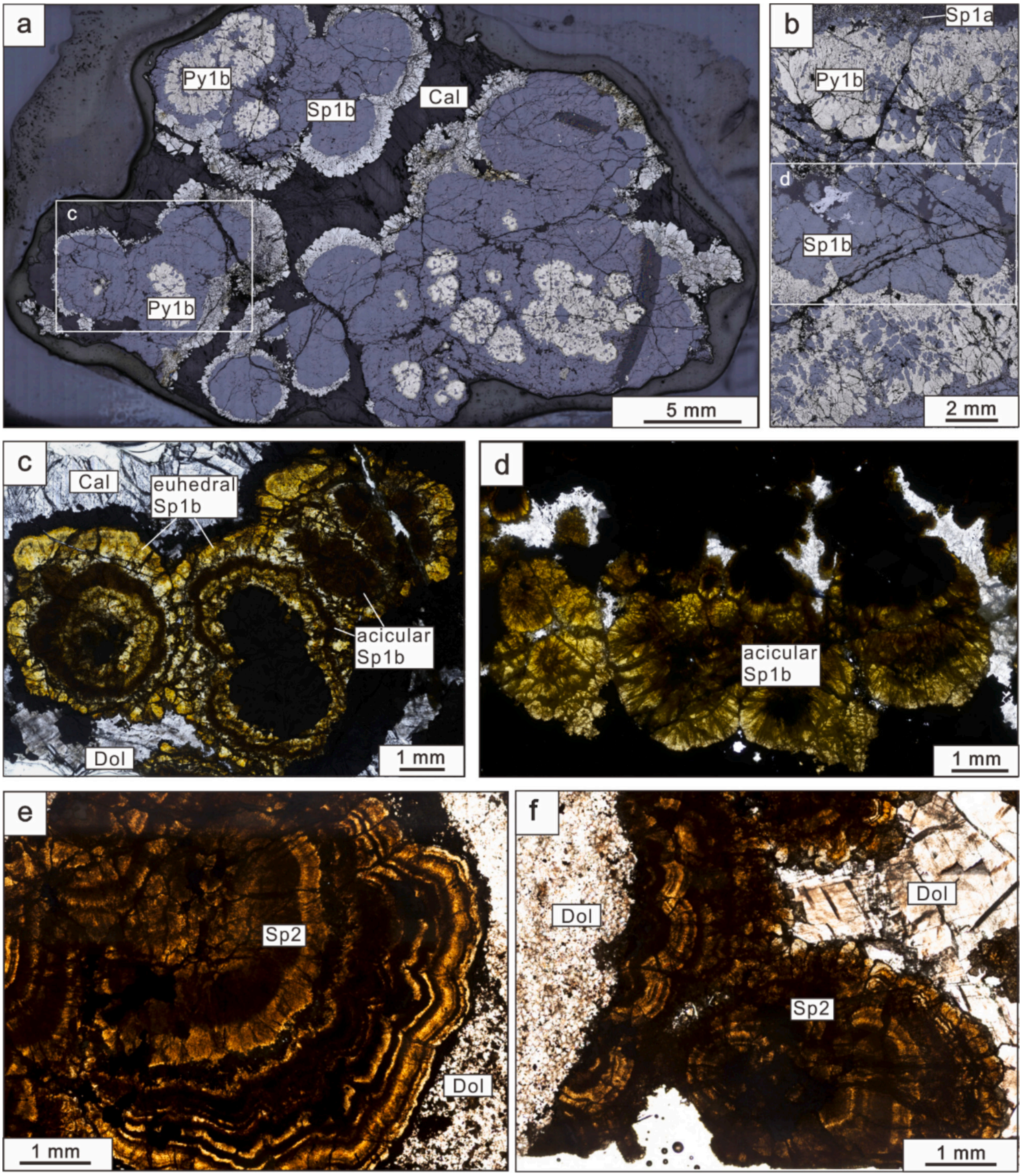

Fig. 4. Representative photo-micrographs showing textures of sulfides. (a) Spheroidal Sp1b and Py1b. (b) Intergrown Sp1b and Py1b. (c) The dark Sp1b is acicular, whereas the light Sp1b is euhedral. (d) Acicular Sp1b. (e-f) Sp2 showing color zonings.

(Fig. 4c, d). Marcasite occurs as acicular crystals intergrowing with euhedral pyrite aggregates (Fig. 4c).

Type 2 ores are characterized by yellow to brown-colored sphalerite (Sp2) (Fig. 3e-f), along with minor pyrite (Py2) and galena with calcite. Sp2 typically forms veinlets along fractures or occurs as globular aggregates (Fig. 3e, f). More complex color-zoning textures are observed in Sp2 compared to Sp1 (Fig. 4e, f). The dark-brown and light-yellow zonings have both acicular textures (Fig. 4e, f). Type 2 ores contain angular and sharp-edged Type 1 ore breccias (Sun et al., 2024), indicating that Type 1 ores are early mineralization and the Type 2 ores represent the later stage of mineralization.

5.2. *Minor and trace elements of sulfides*

LA-ICP-MS results of sphalerite include 4 analyses of Sp1a, 23 analyses of Sp1b, and 13 analyses of Sp2 (ESM 2).

Disseminated Sp1a contains relatively high Fe, Cd, Ge, and Pb concentrations with 10,407–22,006 ppm, 469–760 ppm, 463–692 ppm, and

582–957 ppm, respectively. Detectable Mn and Tl contents are also detected in Sp1a (43.8–66.8 ppm and 16.9–27.9 ppm, respectively). Concentrations of Cu, Ga, Ag, and Sn are below 10 ppm (2.16–5.31 ppm, 1.23–17.8 ppm, 0.68–1.70 ppm, and 6.44–9.19 ppm).

In Sp1b, Fe, Cd, and Pb concentrations are more variable compared with Sp1a (Fig. 5), containing 765–41,183 ppm Fe, 77.0–4,242 ppm Cd, and 206–6,198 ppm Pb (ESM 1). In Sp1b, Ge concentrations are higher than in Sp1a (Fig. 5) and range between 221 and 1,916 ppm with a median value of 891 ppm. Concentrations of Mn, Tl, Cu, Ga, Ag, and Sn are 12.2–236 ppm, 3.75–94.3 ppm, <0.38–4.21 ppm, 1.52–227 ppm, 0.13–6.92 ppm, and 1.78–26.4 ppm, respectively.

Compared with Sp1a and Sp1b, Sp2 exhibits similar concentrations of Mn (70.2 to 122 ppm), Fe (6,594 to 16,798 ppm), Cu (<0.37 to 5.31 ppm), Cd (292 to 1,950 ppm), Tl (14.1 to 31.1 ppm), and Pb (1,379 to 4,345 ppm), but lower concentrations of Ga (3.16 to 24.6 ppm) and Ag (<0.02 to 0.15 ppm). In Sp2, Ge concentrations range between 535 and 931 ppm, with a median value of 752 ppm (Fig. 5).

5.3. In situ sulfur isotopic compositions of sulfides

The in situ $\delta^{34}S_{CDT}$ values of sulfides are listed in ESM 2 and are presented in Figs. 6–9. The $\delta^{34}S_{CDT}$ values of all sulfide minerals are between −13.2 and − 1.0 ‰ (mean − 6.9 ‰). Both Py1a and Py1b have variable $\delta^{34}S_{CDT}$ values (−10.6 to −2.3, −13.7 to −4.8, respectively). Sp1a and Sp1b have $\delta^{34}S_{CDT}$ values ranging from −7.0 to −4.9 ‰ (mean − 5.9 ‰) and − 13.2 and − 1.0 ‰ (mean − 6.4 ‰), respectively. At the center of the vein, Sp1b has more negative $\delta^{34}S_{CDT}$ values (Fig. 7c and 8 c). The $\delta^{34}S_{CDT}$ values of Sp2 are −10.4 ‰ and − 3.8 ‰ (mean − 7.7 ‰). Sp1b has more variable $\delta^{34}S_{CDT}$ values than other generations of sphalerite (Fig. 6a).

5.4. In situ Pb isotopic compositions of sulfides

The Pb isotopic ratios of sulfides from the Banbianjie deposit are listed in ESM 2 and are shown in Figs. 6–9. The $^{206}Pb/^{204}Pb$, $^{207}Pb/^{204}Pb$, and $^{208}Pb/^{204}Pb$ ratios of Py1b range from 18.330 to 18.353, 15.725 to 15.735, and 38.288 to 38.326, respectively. The $^{206}Pb/^{204}Pb$, $^{207}Pb/^{204}Pb$, and $^{208}Pb/^{204}Pb$ ratios of Sp1a, Sp1b, and Sp2 change in the ranges of 18.337–18.354, 15.724–15.738, and 38.290–39.338, respectively. The Pb isotopic compositions of the sulfides are relatively constant. Spatially, the Pb isotopic compositions of Sp1b and Sp2 are consistent within the range of error (Figs. 7d, 8d, and 9d).

6. Discussion

6.1. Lead and reduced sulfur sources

The narrow range of Pb isotopic ratios ($^{206}Pb/^{204}Pb$, 18.337–18.354; $^{207}Pb/^{204}Pb$, 15.724–15.738) indicates that the metals are derived from the upper crust (Fig. 6b) and suggests a relatively uniform crustal source (Zartman and Doe, 1981). In Fig. 6b, the Pb isotopic ratios of Py and Sp from the Banbianjie deposit are plotted with the data from the Wuxun (Balang) Formation, which comprises shales with limestones (Wang and Shi, 1997). Furthermore, in the Wuxun Formation, high contents of Pb and Zn contents (23.6 ppm and 98.6 ppm, respectively, Wang and Shi, 1997) are reported and suggest that the Wuxun Formation is likely to be the source of Pb and other ore-forming metals.

This study provides a comprehensive overview of the Ge contents and Pb isotopic compositions in carbonate-hosted Pb-Zn deposits within the southwestern Yangtze block (Fig. 6c-d). Deposits with high Ge concentrations (Ge ≥ 100 ppm) are predominantly located in the left segment of the basement rocks, the lower sections of the Ediacaran-Cambrian sedimentary rocks, and the central portions of the Devonian-Permian sedimentary rocks. Because there are some spatial overlaps between Ge-rich and Ge-poor deposits (Fig. 6c), the precise origins of Ge remain ambiguous. However, the trend between Ge concentrations and the $^{206}Pb/^{204}Pb$ ratio (Fig. 6d) suggests that the Ge-rich deposits may derive from rocks with more stable lead isotopic signatures.

Sulfur isotopic values of sphalerite (−11.9 ‰ to −1.01 ‰) and pyrite (−13.7 ‰ to −2.3 ‰) in the Banbianjie deposit are significantly lower than those of Devonian seawater sulfate and evaporites (19–27 ‰) (Fig. 6a). The $\delta^{34}S$ values of sulfides suggest that bacterial sulfate reduction (BSR) is likely to be the dominant mechanism producing reduced sulfur (Leach et al., 2005, 2010). Because the formation temperatures of sulfides are mainly above 100 °C, BSR may have occurred prior to the sulfide deposition (Leach et al., 2005, 2010; Doran et al., 2022).

6.2. Formation temperatures

The formation temperature of Sp1b was estimated by applying the sulfur isotope geothermometer. The difference in $\delta^{34}S$ values among sulfides can estimate their temperature of equilibration (Ohmoto and Rye, 1979). The $\delta^{34}S$ values of coexisting Py1b and Sp1b of the Banbianjie deposit are shown in ESM 1. In the sample BBJ-1, $\delta^{34}S_{FeS} > \delta^{34}S_{ZnS}$ is observed for pyrite-sphalerite pairs, suggesting a possible equilibrium between the Sp1b and Py1b (Ohmoto and Rye, 1979). However, in sample BBJ-5, $\delta^{34}S_{FeS} < \delta^{34}S_{ZnS}$ is identified for coexisting Py1b and Sp1b. Therefore, we have only selected the sample BBJ-1 for

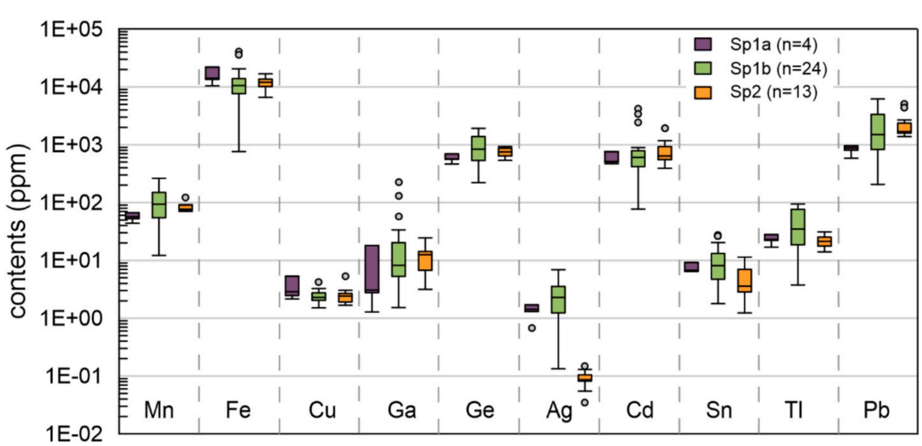

Fig. 5. Box plot showing trace element variations of Sp1a, Sp1b, and Sp2. Abbreviations: Sp1a, sphalerite 1a; Sp1b, sphalerite 1b; Sp2, sphalerite 2.

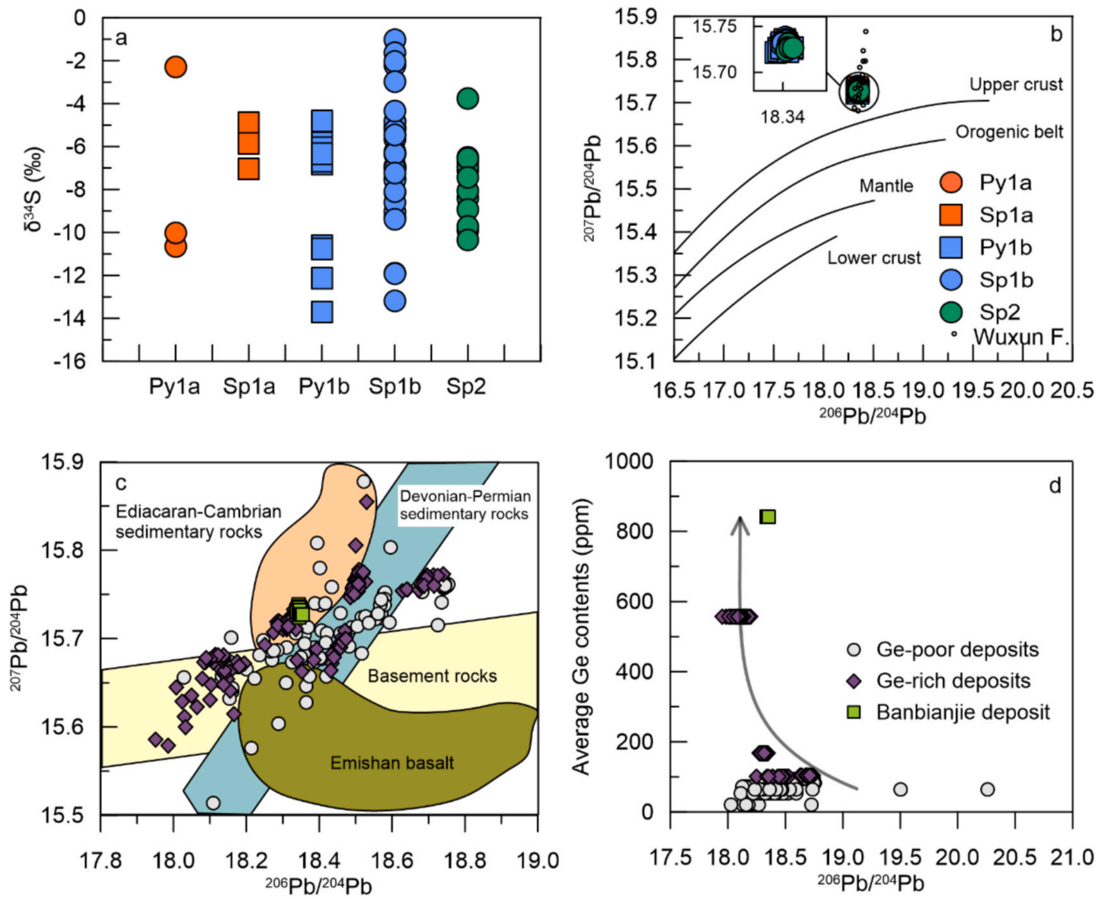

Fig. 6. In situ S-Pb isotopic values of sphalerite and pyrite from the Banbianjie deposit. (a) In situ S isotopic values of different types of sulfides. (b) In situ Pb isotopic ratios of different types of sulfides (after Zartman and Doe, 1981). The Pb isotopic ratios of Wuxun Formation are from Wang and Shi (1997).

the following estimation of ore-forming temperature.

The $\Delta^{34}S$ values of pyrite-sphalerite are grouped in two intervals in the sample BBJ-1: 0.6 ‰–0.7 ‰ and 1.2 ‰–1.8 ‰, respectively. The sulfur isotope geothermometer estimates formation temperatures ranging from 417 to 446 °C and from 136 to 224 °C, respectively. The low temperatures (136–224 °C) interval is comparable to the mineralization temperatures (<250 °C) of carbonate-hosted deposits (Leach et al., 2005, 2010; Zhuang et al., 2019), but the high temperature (417–446 °C) interval is much higher than those of carbonate-hosted deposits. We suggest that the higher temperatures (417–446 °C) are unrealistic data due to sulfide pairs being formed in disequilibrium (Ohmoto and Rye, 1979).

6.3. Changes of physical-chemical conditions

Previous studies have documented that the enrichment of Ge in sphalerite may result from internal mechanisms characterized by self-organization (Sun et al., 2024). However, the external factors, such as fluid sources, physical-chemical conditions (temperature, pH, and redox state), and syntectonic recrystallization, may also influence the Ge enrichment (Cook, 1996; Barrie et al., 2009; Cook et al., 2009; Gagnevin et al., 2014; Bauer et al., 2019; Cugerone et al., 2020; Yang et al., 2025).

6.3.1. Formation temperatures

It is difficult to be certain that the trace element zoning in sphalerite reflects changes of temperature. The relationship between trace element concentrations and temperatures is complex (Scott, 1973; Keith et al., 2014). For example, some authors suggest that Cd contents decrease with increasing temperatures (Liu et al., 1984). There is also evidence that Cd is observed up to 20 mol% at high temperatures (Tauson et al., 1977).

Recently, some studies have suggested the formation temperatures by GGIMFis geothermometer, which is based on an empirical relationship between the composition of sphalerite and fluid inclusion microthermometry (Frenzel et al., 2016). The estimation mainly works for individual deposits or distinct mineralization events within deposits (Frenzel et al., 2016; Bauer et al., 2019; Sun et al., 2021) and may not be efficient for different stages of mineralization within an individual mineralization event. In the Banbianjie region, the contents of Ge exhibit a positive correlation with those of Fe (Figs. 7–9). This observation contrasts with the empirical formula of the GGIMFis geothermometer. Consequently, employing this geothermometer requires careful consideration of the interrelationships among trace elements at the scale of studies.

6.3.2. pH

The most important factors that control the solubility of metals in MVT ore-forming fluids are pH, temperature, and reduced sulfur (Leach et al., 2005; Zhou et al., 2021b). The mineral assemblages of the Banbianjie deposit suggest a possible occurrence of pH changes. Pyrite-marcasite is a mineral pair sensitive to the pH of hydrothermal fluids. It is well documented that pyrite dominantly forms from mildly acidic to alkaline fluids (pH > 4.5), whereas marcasite only precipitates from acidic fluids (pH < 4.5) (Allen and Crenshaw, 1914; Schoonen and Barnes, 1991). Thermodynamic modeling suggests that sphalerite may precipitate at higher pH conditions than pyrite (Xie et al., 2020). Combining the mineralogical observation in this case, the colloform texture in stage 1 veins, which shows a paragenesis of pyrite→sphalerite→pyrite→sphalerite→pyrite→marcasite, we can infer that the ore-forming fluids were initially weakly acidic and went through several

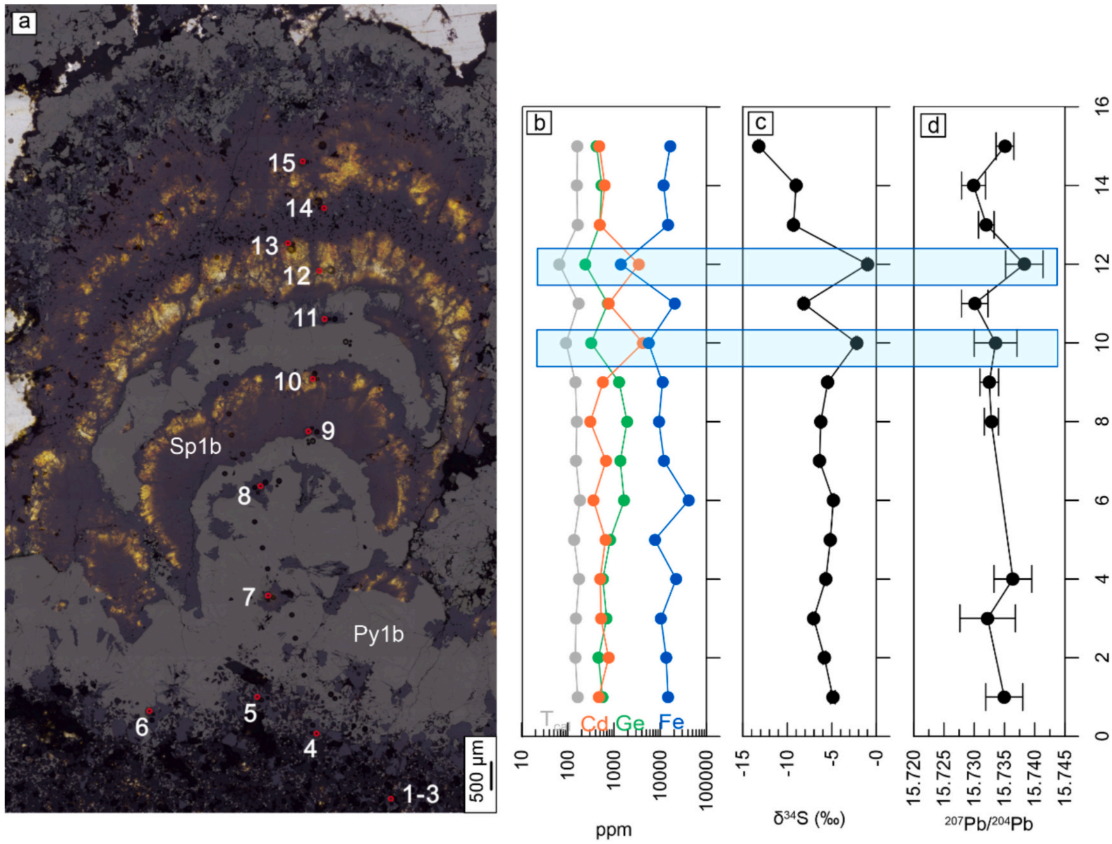

Fig. 7. Transect showing $^{207}Pb/^{204}Pb$, $\delta^{34}S$, and trace element variation in Sp1 from the Banbianjie deposit. (a) Location of analysis points (modified from Sun et al., 2024). (b) $^{207}Pb/^{204}Pb$, $\delta^{34}S$ and trace element variation in Sp1. The blue rectangles indicate subordinate fluids. Abbreviations: Py1b, pyrite 1b; Sp1b, sphalerite 1b. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

rounds of buffering processes. As Ge is mainly hosted in sphalerite, not pyrite (Frenzel et al., 2014), pH changes may have controlled the Ge enrichment via its constraint on sulfide phases.

6.3.3. Redox state

Sulfides are dominantly S-bearing minerals and minor barite precipitates after the precipitation of pyrite and sphalerite. Therefore, the hydrothermal system seems to have evolved from a reduced state to an oxidized state. The observation, coupled with the fact that the outer layers of sphalerite and pyrite have lower $\delta^{34}S$ values (−13.7 ‰ to −10.7 ‰), implies that fluid oxidization may cause the shift to more negative $\delta^{34}S$ values (Fig. 7c). The negative shift is not associated with trace element oscillatory spatially (Fig. 7b, c), suggesting that redox changes may not result in trace element zonation (Ohmoto and Rye, 1979).

6.4. Fluid mixing

In situ Pb isotope analysis reveals that Pb isotope ratios do not show significant spatial variations (Figs. 7–9), indicating that most Pb and Ge may be derived from a single source. Sulfur isotopic compositions show relatively large variations, generally correlating with trace element contents. Sphalerite with $\delta^{34}S$ values ranging from −2.2 ‰ to −1.0 ‰ is enriched in Cd and depleted in Fe, Ge, Mn, and Pb, while sphalerite with $\delta^{34}S$ values between −13.2 ‰ and − 4.3 ‰ shows variable levels of Fe, Ge, Cd, Mn, and Pb (Figs. 7–10). Considering that previous studies have reported the presence of S-Zn-Ge-Pb nanoparticles in sphalerite (Sun et al., 2024), the influence of these nanoparticles on sulfur isotopic fractionation must be taken into account. However, since no correlation is observed between Ge-rich analyses and $\delta^{34}S$ values, this impact may be minimal. Therefore, the distinct trace element contents and $\delta^{34}S$ values likely suggest that there are two types of fluid: 1) predominant fluids with $\delta^{34}S$ values ranging from −11 ‰ to −4 ‰ and normal Cd concentrations; 2) subordinate fluids with $\delta^{34}S$ values from ~1 ‰ and high Cd concentrations (Fig. 10b). The predominant Ge-rich fluids may be the basin brines that extract metals from Wuxun Formation. Previous study suggests that magmatic fluids could be rich in Cd (Fan et al., 2023), however, there is no related magmatism in this region. Therefore, the Cd-rich fluids may be derived from a similar source (Precambrian metamorphic rocks) of the Niujiaotang MVT Zn-Cd deposit (Zhou et al., 2022), which is located to the east of Banbianjie and contains 5300 t Cd with average Cd contents of 1.4 wt% in sphalerite (Ye et al., 2012).

Sun et al. (2024) proposed that the trace element zoning in sphalerite may result from a self-organization process, independent of fluid sources. The $\delta^{34}S$ and LA-ICP-MS analyses were conducted along transects to distinguish the different mechanisms. Two types of Cd anomaly are revealed in Sp1: 1) 1–2 extremely positive Cd anomalies ($2Cd_n/(Cd_{n-1} + Cd_{n+1})$ = 3.1–6.3), coinciding with positive anomalies in $\delta^{34}S$ values (~1 ‰); 2) moderate Cd anomalies ($2Cd_n/(Cd_{n-1} + Cd_{n+1}) < 2.0$) that are decoupled with the S isotopic compositions (Figs. 7–8). The coupled and decoupled characteristics suggest that extreme anomalies may be linked to fluid mixing, while moderate anomalies could be induced by a self-organization process.

The coupled positive anomalies in Cd and $\delta^{34}S$ may be attributed to one to two influxes of Cd-rich fluid mixing into the predominant ore-forming fluids during the crystallization (Figs. 7–9). Furthermore, the mixing of Ge-rich fluids with pulses of Cd-rich fluids could lead to the dilution of Ge concentration, subsequently resulting in Ge-depleted zonings. Another potential mechanism by which the Cd-rich fluid might cause Ge depletion may be the crystallization rate. Luo et al. (2022) established that colloform sphalerite consists of rapid growth zones enriched in Ge and slower growth zones enriched in Cd. Therefore,

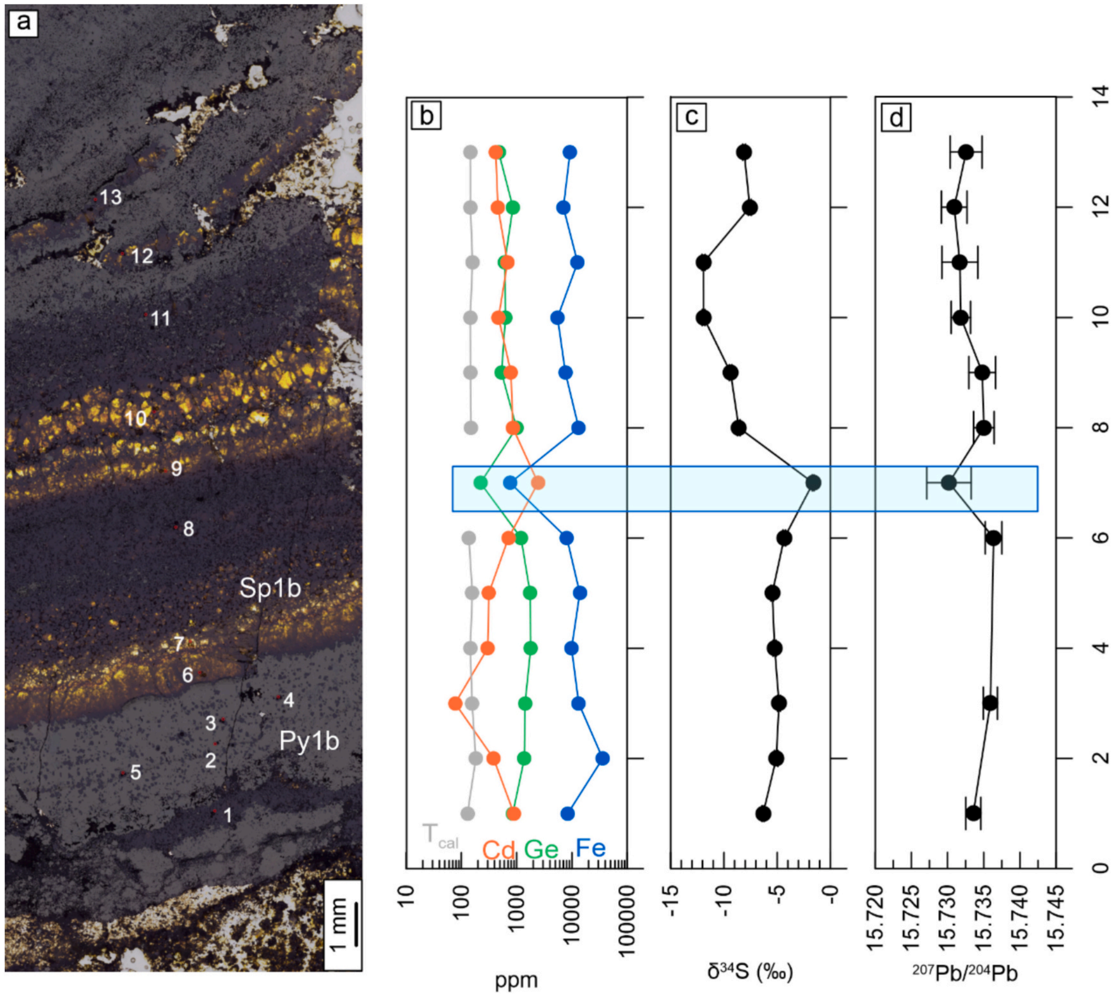

Fig. 8. Transect showing $^{207}Pb/^{204}Pb$, $\delta^{34}S$, and trace element variation in banded Sp1b from the Banbianjie deposit. (a) Location of analysis points (modified from Sun et al., 2024). (b) $^{207}Pb/^{204}Pb$, $\delta^{34}S$ and trace element variation in banded Sp1b. The blue rectangles indicate zones interpreted to be influenced by influxes of subordinate fluids. Abbreviations: Py1b, pyrite 1b; Sp1b, sphalerite 1b. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

fluid mixing with Cd-rich fluids could also induce a slower precipitation rate of colloform sphalerite, potentially leading to inefficient Ge enrichment.

Simulation was further conducted to evaluate the mixing proportion and its influence on Ge contents. The mixing model is based on the two fluid endmembers illustrated by trace elements and sulfur isotope compositions. The mixing curve indicates that the proportion of Cd-rich fluid may be up to 70 % in Sp1, while in Sp2, it is lower (20–30 %) (Fig. 10). In Sp1, the negative impacts of fluid mixing on Ge contents are more pronounced than those of self-organization (Fig. 10a). However, in Sp2, these impacts are comparable to those of self-organization (Fig. 10a).

Previous study demonstrates that the fluid mixing between Ge-bearing reduced fluid and oxidized basinal brine may trigger Ge-oxide mineral precipitation in Huize and Maoping carbonate-hosted deposits (Niu et al., 2024). Their study suggests that the mixing controls the Ge enrichment on the deposit scale. Our in situ analyses further indicate that the fluid mixing may also influence the Ge enrichment at the mineral scale.

7. Conclusions

Colloform sphalerite in the Banbianjie deposit shows high contents of Ge (up to 1916 ppm, in situ measurements). S-Pb isotopic compositions in sphalerite suggest that BSR was likely the main process producing reduced sulfur and the ore-forming metals may be derived from the Wuxun Formation. The distribution of trace elements in colloform sphalerite shows three characteristics, including (1) oscillatory zoning in sphalerite, (2) a negative correlation between Ge and Cd, (3) coupled anomalies in Cd and $\delta^{34}S$, (4) decoupled correlation between Ge contents and S-Pb isotopic values. These observations suggest that local fluid mixing may account for Ge oscillatory zonings in sphalerite, while most of the oscillatory zonings may result from self-organization processes. This study shows that external mechanisms, such as fluid mixing, can significantly influence the Ge contents in colloform sphalerite.

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CRediT authorship contribution statement

Guotao Sun: Writing – original draft, Project administration, Methodology, Investigation. **Jia-Xi Zhou:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Alexandre Cugerone:** Writing – review & editing. **Lingli Zhou:** Writing – review & editing. **Kai Luo:** Investigation. **Maoda Lu:** Resources, Investigation.

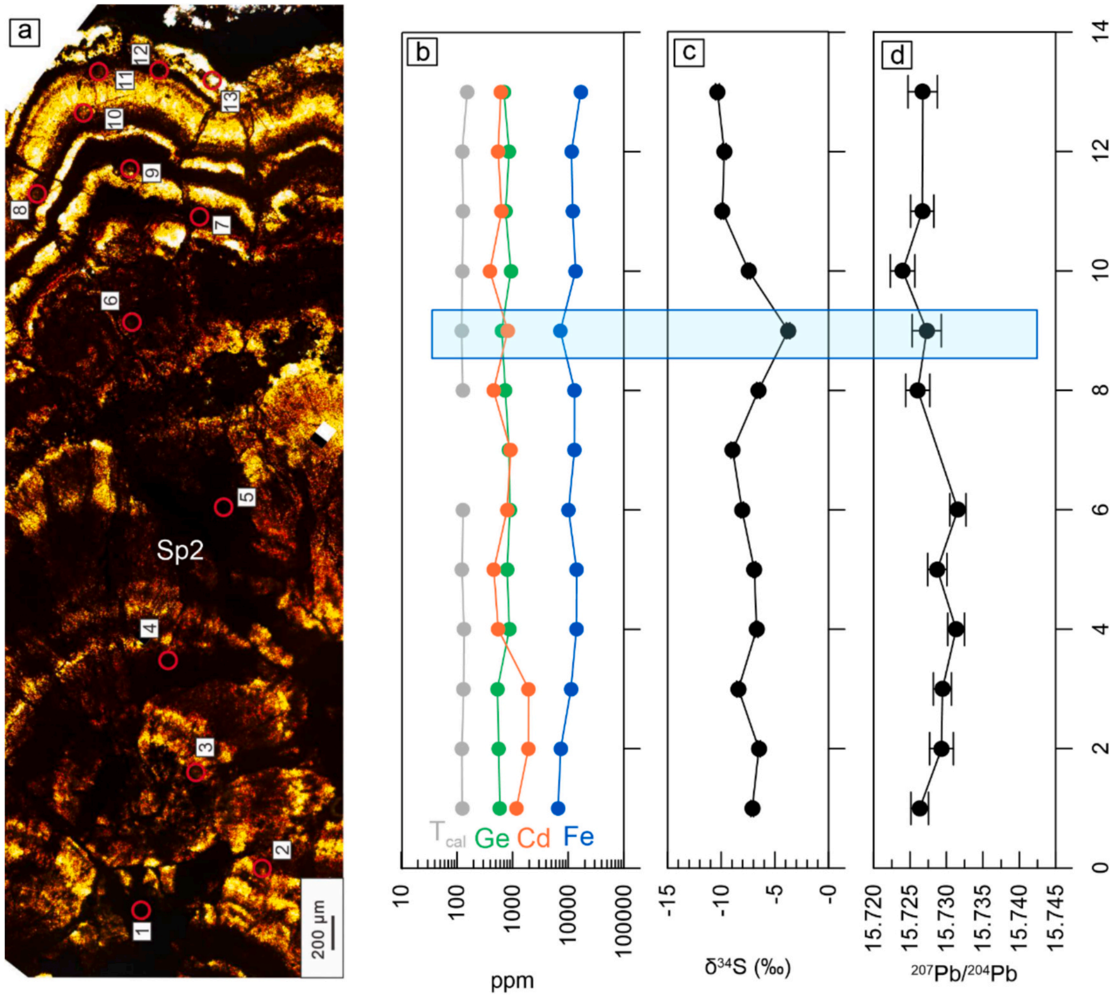

Fig. 9. Transect showing $^{207}Pb/^{204}Pb$, $\delta^{34}S$, and trace element variation in Sp2 from the Banbianjie deposit. (a) Location of analysis points (modified from Sun et al., 2024). (b) $^{207}Pb/^{204}Pb$, $\delta^{34}S$ and trace element variation in Sp2. The blue rectangle indicates subordinate fluids. Abbreviations: Sp2, sphalerite 2. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

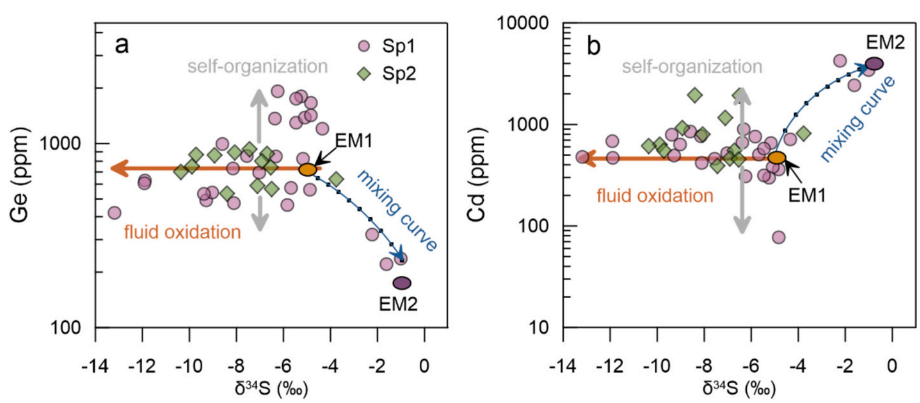

Fig. 10. Binary plots for trace elements versus S isotopic values. (a) Ge versus $\delta^{34}S$, (b) Cd versus $\delta^{34}S$. Endmember 1: Ge 700 ppm, Cd 500 ppm, $\delta^{34}S = -5$ ‰; Endmember 2: Ge 180 ppm, Cd 4200 ppm, $\delta^{34}S = -1$ ‰. Abbreviations: Sp1a, sphalerite 1a; Sp1b, sphalerite 1b; Sp2, sphalerite 2.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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